

Scalability & Future Plan

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Team ID	SWTID-2026-8096
Project Name	Rising Waters
Maximum Marks	1 Mark

Scalability & Future Plan:

This document outlines how the **Rising Waters – AI-Based Flood Prediction System** can be enhanced to support a larger number of users, process more comprehensive rainfall datasets, and incorporate advanced features in future versions. It also presents a roadmap for improving the system's performance, usability, and prediction capabilities beyond the current implementation.

Current System Limitations:

S.No	Limitation	Impact	Priority to Address (High / Medium / Low)
1	Uses historical rainfall data only and does not integrate real-time weather information.	Predictions may not reflect current weather conditions.	High
2	Uses historical rainfall data only and does not integrate real-time weather information.	Additional environmental factors could improve prediction accuracy.	Medium
3	No user authentication or prediction history is available.	Users cannot save or review previous prediction results	Medium

Scalability Plan:

S.No	Scalability Aspect	Current State	Proposed Upgrade / Solution
1	User Load	Supports a moderate number of users through a single Render deployment.	Deploy using scalable cloud infrastructure with load balancing and auto-scaling.
2	Data Storage	Uses locally stored machine learning model files and static data.	Integrate a cloud database such as PostgreSQL or MongoDB for storing prediction history and user data.

3	Performance	Predictions are processed sequentially by the Flask application	Optimize backend processing, implement caching, and improve model inference efficiency.
4	Security	Basic input validation is implemented	Add user authentication, HTTPS enforcement, secure API endpoints, and role-based access control.

Future Roadmap:

Phase	Planned Feature / Enhancement	Target Timeline	Expected Impact
Phase 1	Integrate real-time weather and rainfall APIs.	Short Term	Improve prediction accuracy using live weather data.
Phase 2	Add GIS-based flood risk mapping and visualization.	Medium Term	Enable users to view flood-prone areas on interactive maps.
Phase 3	Develop Android and iOS mobile applications with SMS and email alerts.	Medium Term	Increase accessibility and provide timely flood warnings.
Phase 4	Enhance the prediction model using larger datasets and advanced deep learning techniques.	Long Term	Improve prediction accuracy, scalability, and support for large-scale disaster management.

The current **Rising Waters** system successfully predicts flood occurrence using rainfall data and machine learning. With future enhancements such as real-time weather integration, GIS mapping, mobile application support, cloud databases, and advanced prediction models, the system can evolve into a comprehensive flood monitoring and early warning platform capable of supporting larger user bases and more accurate disaster management.